



RoadSide

Hold your phone near your bike. It listens to the noise, looks at the chain, and tells you what may be wrong and what to do.

Fully on-device

Works in airplane mode

Built on OnePlus 13R

iQOO Hackathon · Open Innovation

A new noise, and no way to know what it means

Every rider knows the moment: a rattle, click or squeal appears out of nowhere.



You can't search a sound

Is it a dirty chain, worn sprocket teeth, or nothing at all? Typing "weird rattle" into a search box does not help.



Breakdowns happen off-grid

The problem shows up on the highway or a village road — exactly where there is no signal for online help.



Paying just to find out

The fallback is ignoring it and risking a snapped chain, or a mechanic visit for work the bike may not need.

Your phone becomes a roadside check

Three steps, about a minute, nothing leaves the phone.



1. Listen

Record the noise for a few seconds while it happens.



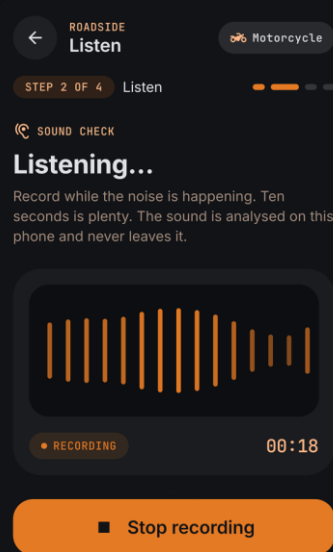
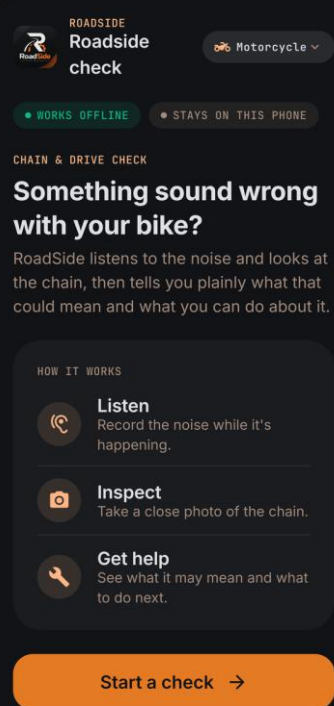
2. Inspect

Take a close photo of the chain and sprocket.



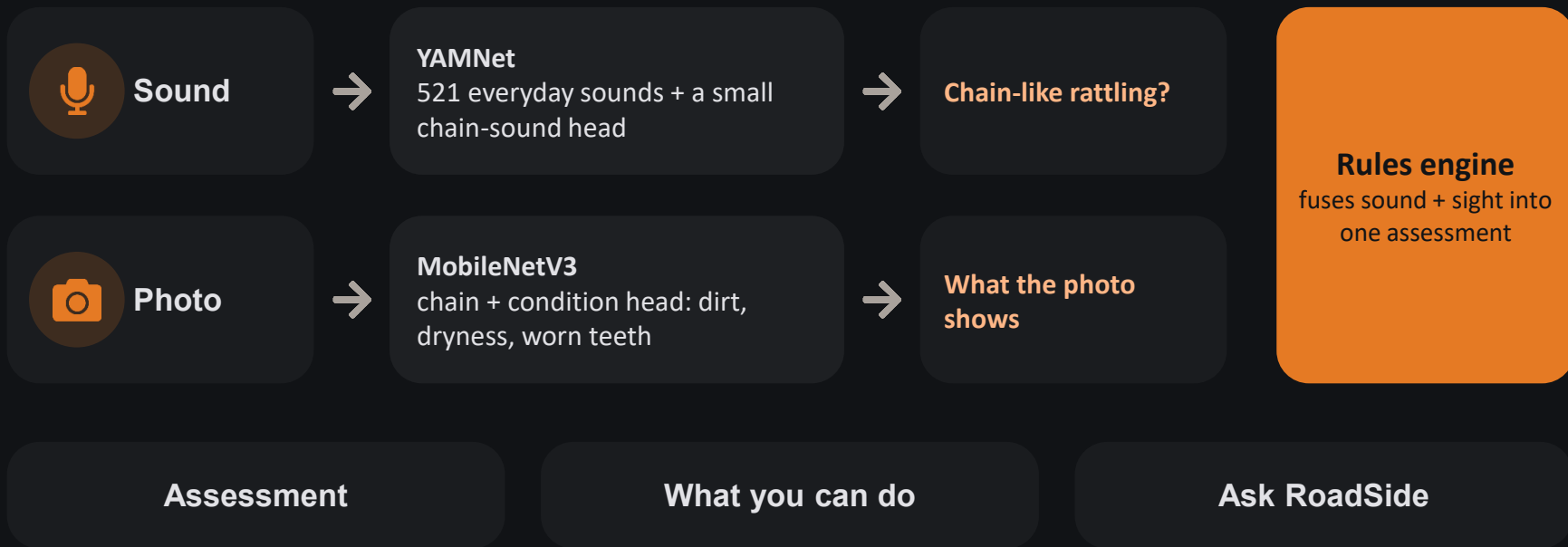
3. Get help

What RoadSide found, what it may mean, what to do — and a safety note.



How it works — two senses, one plain answer

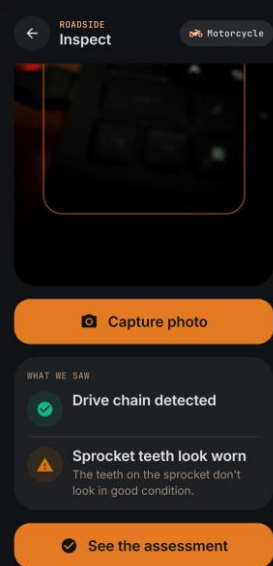
Open pretrained models with small trained heads, all running on the handset.



No internet permission · works in airplane mode · about 37 MB of models on the phone

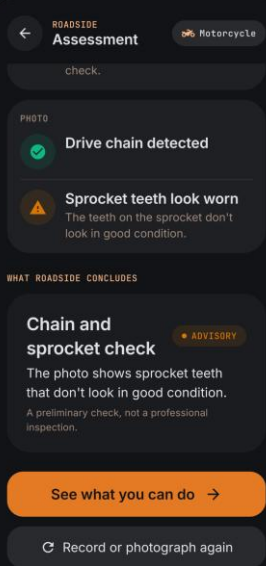
From photo to plain advice

Real screens from the app on a OnePlus 13R.



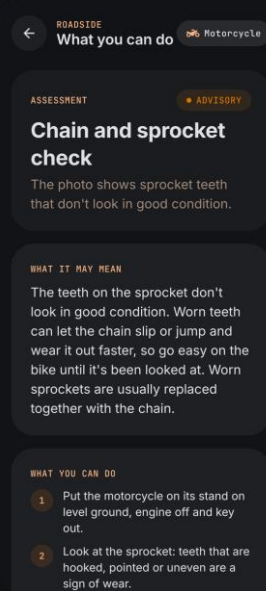
Inspect

Chain detected; sprocket teeth look worn



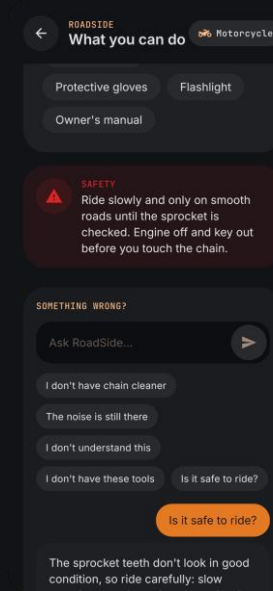
Assessment

Observations kept separate from the conclusion



What you can do

What it may mean, actions, tools, safety



Ask RoadSide

"Is it safe to ride?" answered in context

Honest by design

A wrong "your chain is fine" is worse than "not sure". RoadSide is built to say so.



Observations ≠ conclusions

What the camera saw and what RoadSide concludes are shown separately.



"Not clear enough" beats a guess

Low-confidence photos and sounds return an honest unknown, with a tip to retake.



No invented numbers

No fake Hz, millimetres, torque or percentages anywhere a rider reads.



Your words never change the result

Only the microphone and camera decide; typed text is context for you.

Built and measured, not mocked

Everything below was measured on a OnePlus 13R.

29 / 29

on-device tests pass —
online and in airplane
mode

0

internet permissions —
the app cannot phone
home

~0.1 s

to analyse a photo on the
phone

< 0.5 s

to analyse a 15-second
recording

Also verified: smooth animations (median frame 9 ms), recorder and camera release cleanly, and inference never blocks the screen.

Two-wheelers today, cars next

The pipeline is general; what grows is the set of labelled faults.



Today — working prototype

- Motorcycle chain & sprocket from a photo: chain detected, dirt, dryness, worn teeth
- Chain-like rattling from sound
- Early flags for brake squeal and starter clicking from sound
- Plain-language solution + Ask RoadSide, fully offline



Next — same pipeline, more vehicles

- Cars: engine knock, belt squeal, brake wear, battery and starter faults
- Scooters and more motorcycles, tyres and brakes by photo
- Labelled recordings and photos from many vehicles to train on
- An on-device language model for richer Ask RoadSide answers



Ride smarter.

A roadside check that listens, looks and explains — on the phone you already carry.